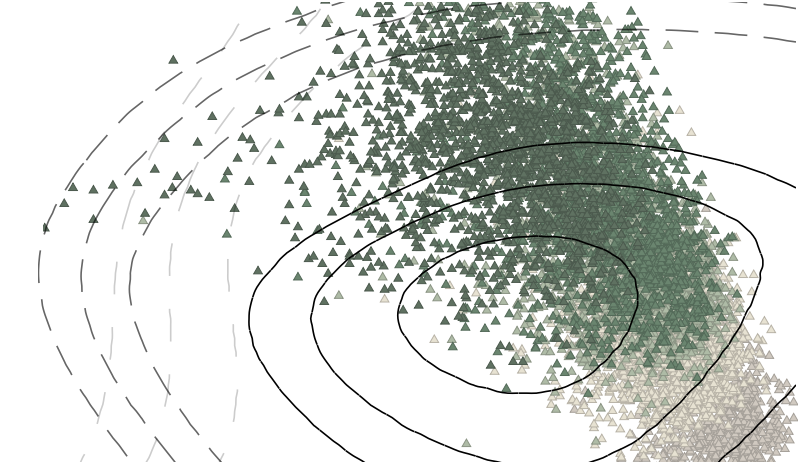
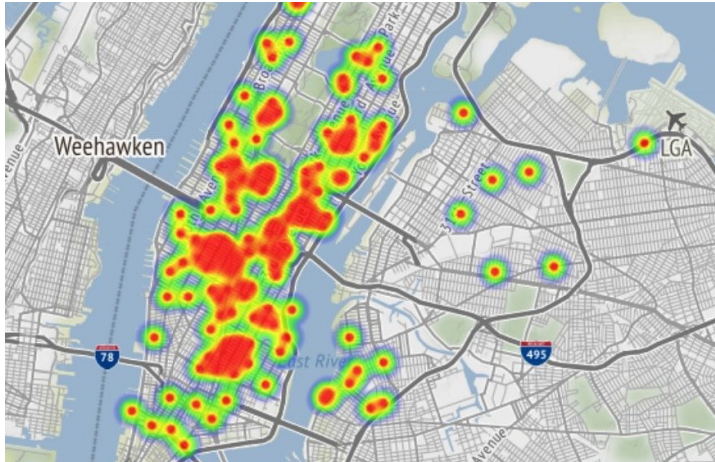
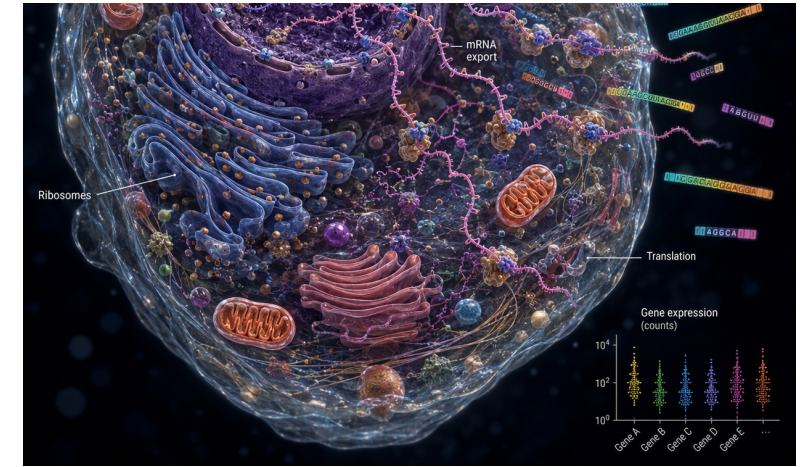
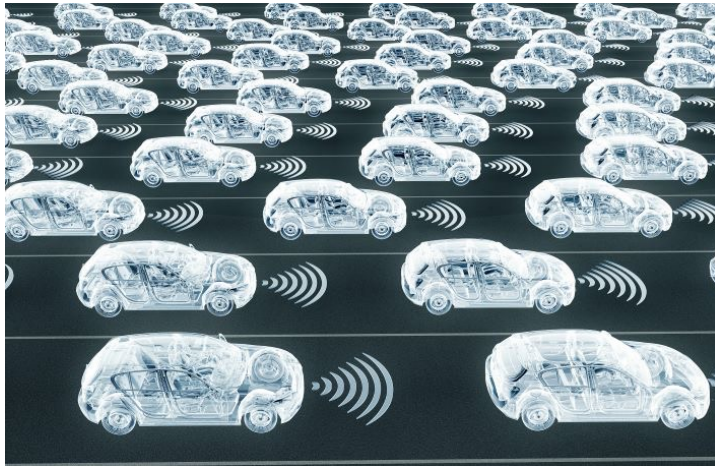


separation principles via optimal transport

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*Sorry! Nicolas will present on Friday the variational structure of the Wasserstein space

real world complexity demands shortcuts





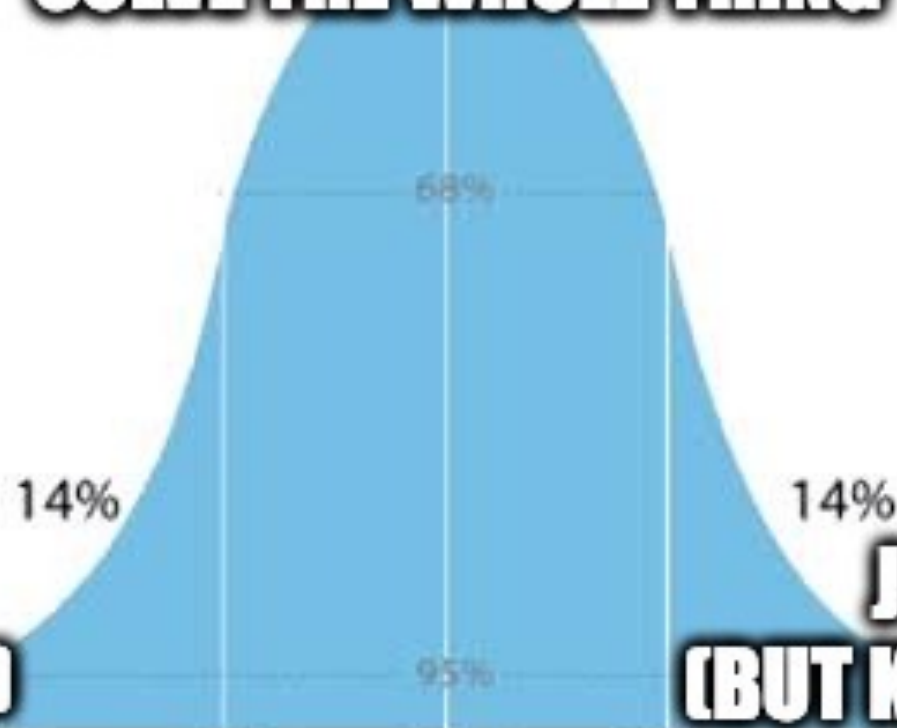
SOLVE THE WHOLE THING



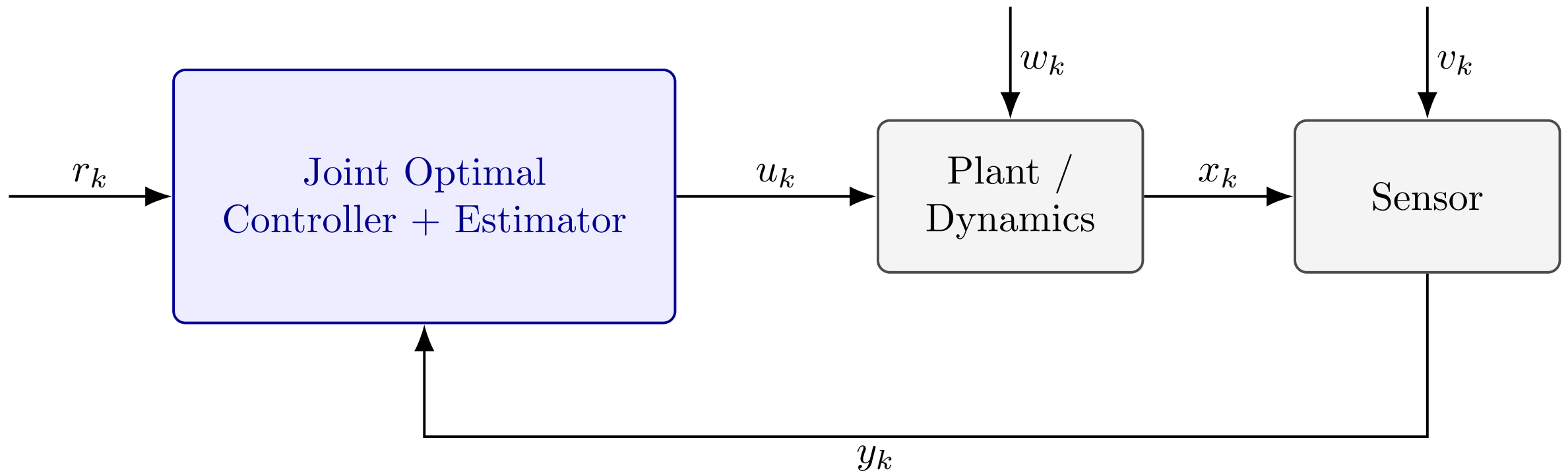
**JUST SEPARATE IT
(SOMETIMES IT WORKS)**



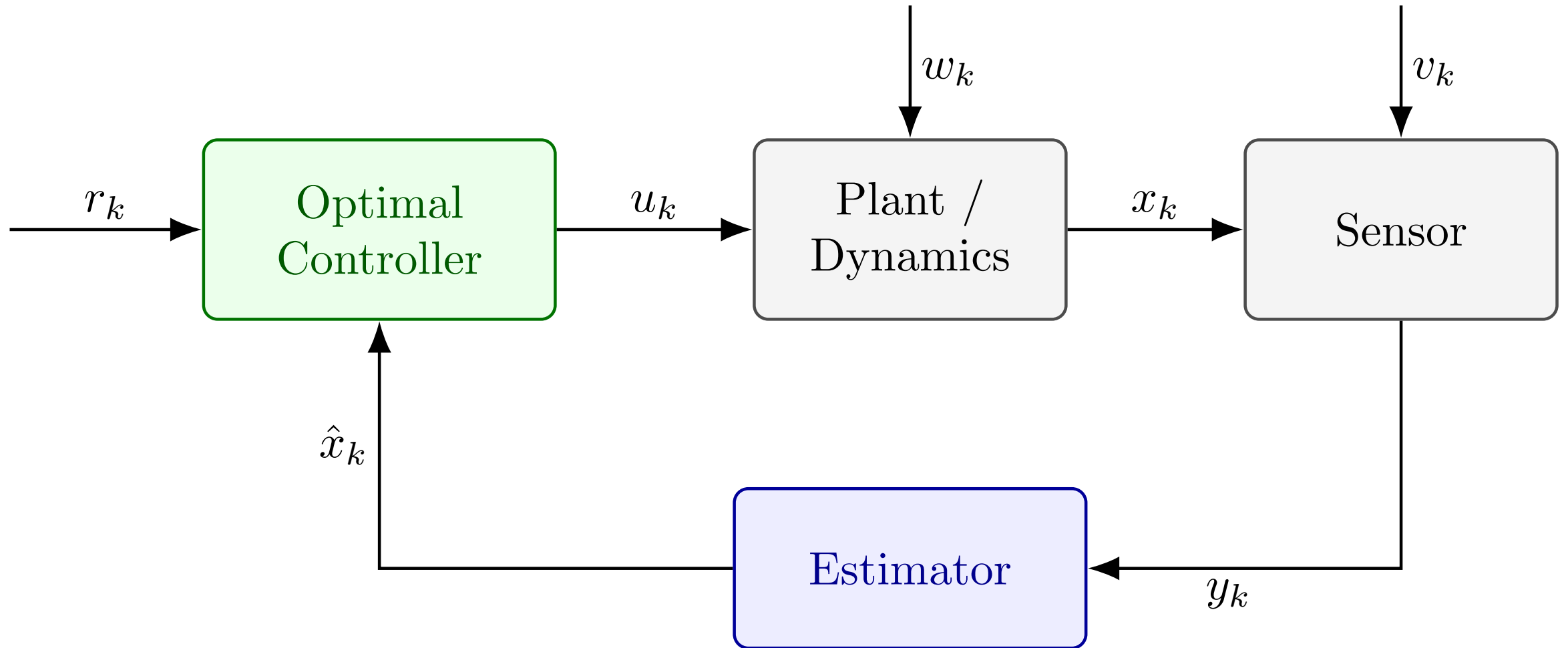
**JUST SEPARATE IT
(BUT KNOW WHY, THIS TALK)**



prominent example: LQG

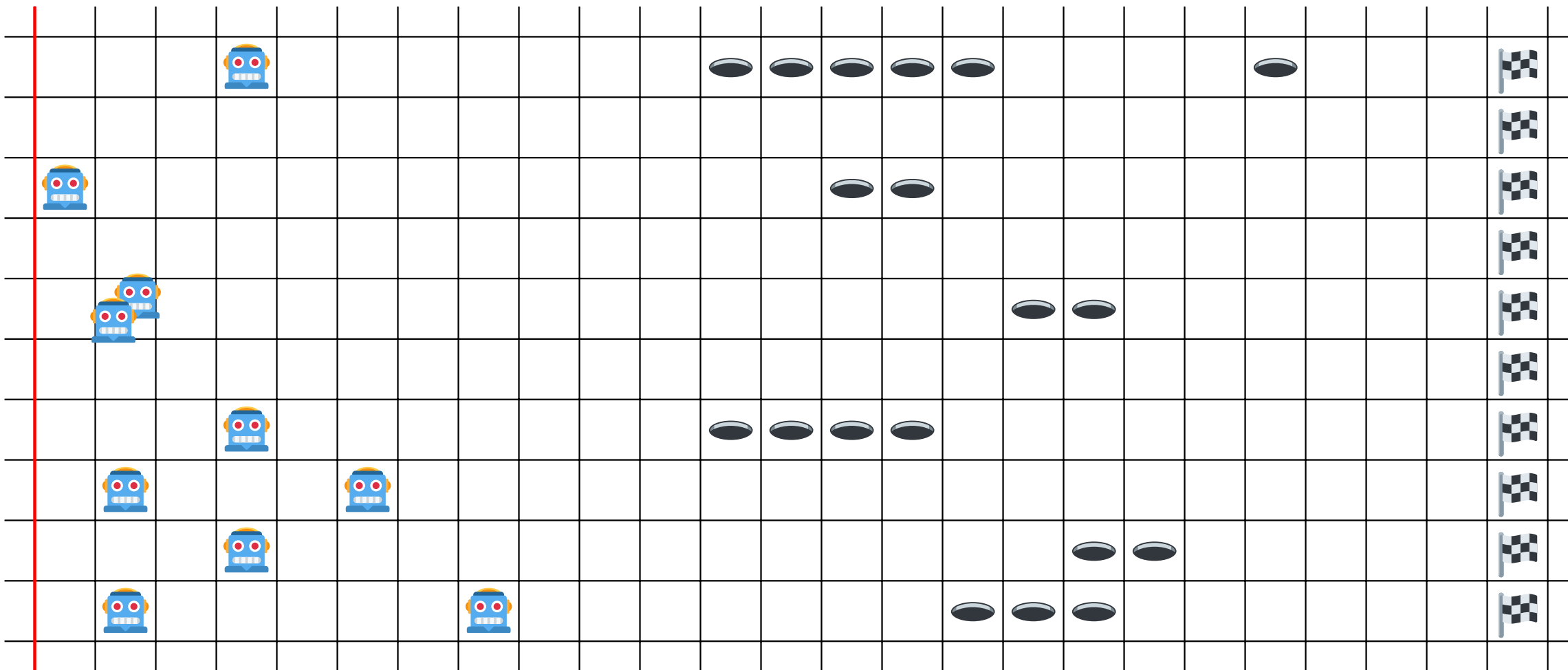


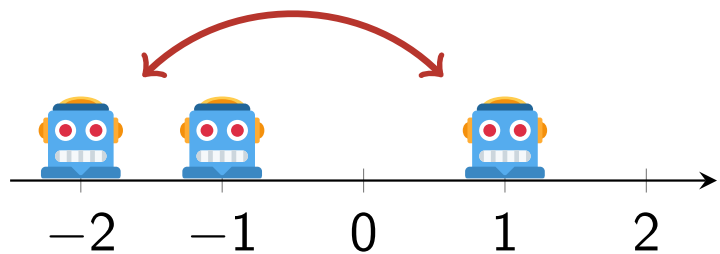
prominent example: LQG





a separation principle for multi-agent control



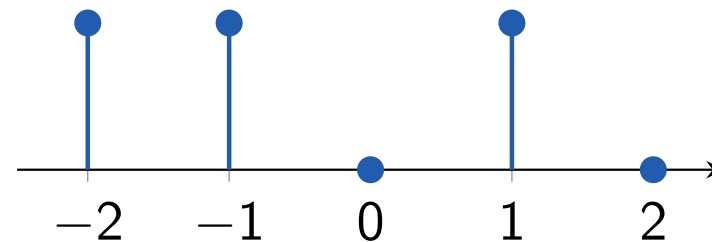
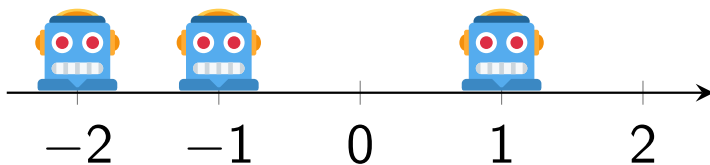


$$x = \begin{bmatrix} \text{state agent 1} \\ \text{state agent 2} \\ \text{state agent 3} \end{bmatrix} \neq \begin{bmatrix} \text{state agent 1} \\ \text{state agent 3} \\ \text{state agent 2} \end{bmatrix}$$

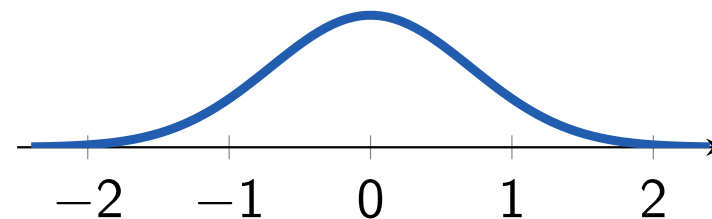
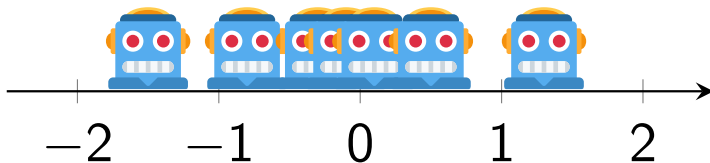
... and this happens because we **artificially** label the agents.

The remedy: Model a fleet **via a distribution**!

Finite fleet



Infinite fleet



Terminal cost (formation cost):

optimal transport discrepancy with transportation cost $g_N : X_N \times R \rightarrow \mathbb{R}_{\geq 0}$ from a reference ρ

Stage cost (aggregate cost):

average cost over the fleet

$$g : X_k \times U_k \rightarrow \mathbb{R}_{\geq 0}$$

$$J(\mu, \rho) = \inf_{\substack{\mu_k \in \mathcal{P}(X_k) \\ \lambda_k \in \mathcal{P}(X_k \times U_k)}} \mathcal{K}[g_N](\mu_N, \rho) + \sum_{k=0}^{N-1} \mathbb{E}^{\lambda_k} [g_k]$$

State distribution μ_k :

distribution of agents

State-input distribution λ_k :

probability of being at a given state and use a given action

s.t.

$$\mu_{k+1} = f_{k\#} \lambda_k,$$

$$\mu_0 = \mu,$$

$$(\text{proj}_{X_k}^{X_k \times U_k})_{\#} \lambda_k = \mu_k.$$

Dynamics:

$f_k : X_k \times U_k \rightarrow X_{k+1}$
dynamics of each agent

Challenge:

We can use dynamic programming,
but the cost-to-go and feedback laws are **arbitrary functions** over distributions.

Theorem: A separation principle holds true:

1. assign agents to targets solving a fleet-level dispatching problem;
2. route individual agents.

For the optimal transport aficionados: At every stage k :

1. The cost-to-go is an optimal transport problem:

$$J_k(\mu_k, \rho) = \mathcal{K}[j_k](\mu_k, \rho) = \inf_{\gamma \in \Gamma(\mu_k, \rho)} \int_{X_k \times R} j_k(x_k, r) d\gamma(x_k, r),$$

where j_k is a single-agent cost-to-go:

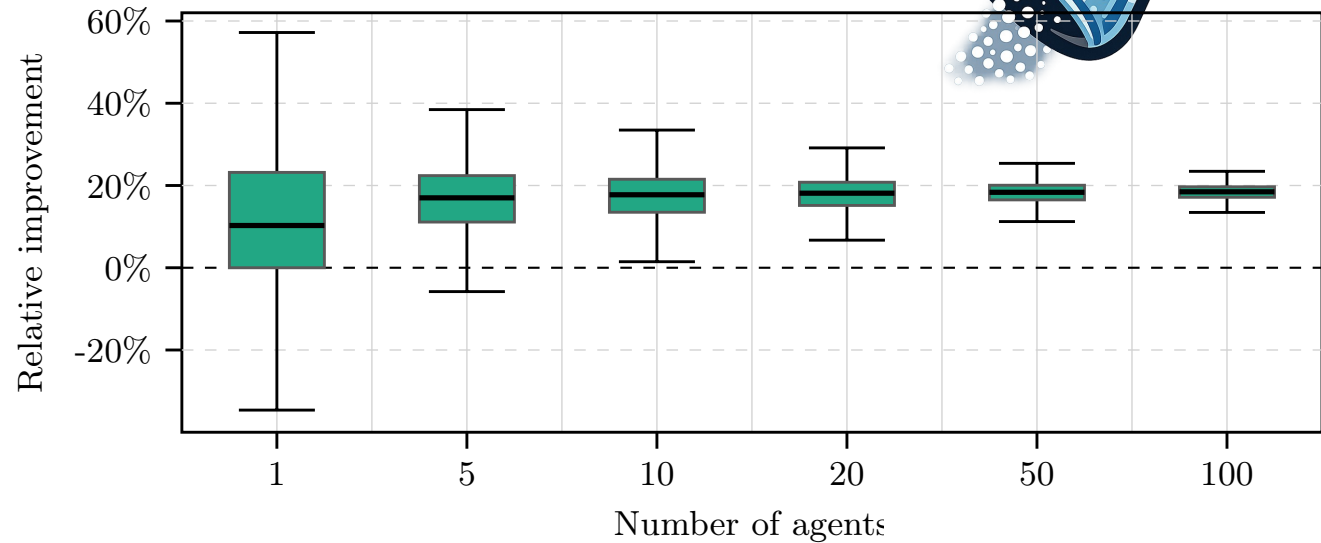
$$j_N(x_N, r) := g_N(x_N, r); \quad j_k(x_k, r) := \inf_{u_k \in U_k} g_k(x_k, u_k) + j_{k+1}(f_k(x_k, u_k), r).$$

2. Optimal state-input distribution:

$$\lambda_k^* = \left(\text{proj}_{X_k}^{X_k \times R}, u_k^* \right)_{\#} \gamma_k^*$$

where u_k^* is the optimal single-agent control input and γ_k^* is the optimal dispatching plan in the cost-to-go $J_k(\mu_k, \rho)$.

... with consequences for learning



Algorithm 1 Separation-based Assignment and Learning via optimal Transport

- 1: **for** each episode **do**
 - 2: Sample starting states and targets.
 - 3: **Match** agents to targets via optimal transport.
 - 4: **Act**: roll out the shared policy with assignments fixed.
 - 5: **Learn**: update the shared policy and value from pooled trajectories.
 - 6: **end for**
-





a separation principle for multi-agent learning

Ribosomes

mRNA
export

Translation



diffusion processes in Switzerland



goal: identify the population dynamics from population data

find the energy functional that best explains the data

$$\mu_{t+1} = \operatorname{argmin}_{\mu \in \mathcal{P}(\mathbb{R}^d)} J(\mu) + \frac{1}{2\tau} W_2(\mu, \mu_t)^2$$

population data at time t+1 (given)

Wasserstein distance

discretization

population distribution at time t (given)

$$J(\mu) = \int_{\mathbb{R}^d} V(x) d\mu(x) + \beta \int_{\mathbb{R}^d} \log(\rho(x)) \rho(x) dx \quad \Leftrightarrow \quad dX_t = -\nabla V(X) dt + \sqrt{2\beta} dW_t$$

first-order
optimality
conditions
in the
Wasserstein
space

$$\mu_{t+1} = \operatorname{argmin}_{\mu \in \mathcal{P}(\mathbb{R}^d)} J(\mu) + \frac{1}{2\tau} W_2(\mu, \mu_t)^2$$

optimal coupling
between and
(only data-dependent)

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} \left\| \nabla J(\mu_{t+1})(x_{t+1}) + \frac{1}{\tau} (x_{t+1} - x_t) \right\|^2 d\gamma_t(x_t, x_{t+1}) = 0$$

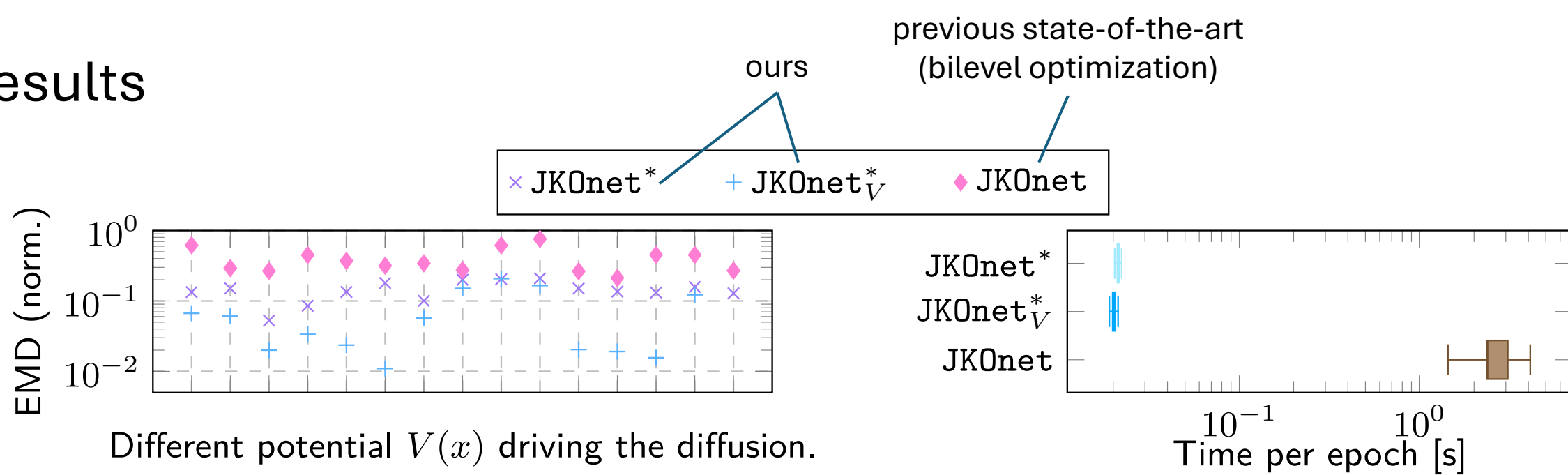
loss
function for
estimation

$$\min_{\theta} \sum_{t=0}^{T-1} \int_{\mathbb{R}^d \times \mathbb{R}^d} \left\| \nabla J_{\theta}(\mu_{t+1})(x_{t+1}) + \frac{1}{\tau} (x_{t+1} - x_t) \right\|^2 d\gamma_t(x_t, x_{t+1}) = 0$$

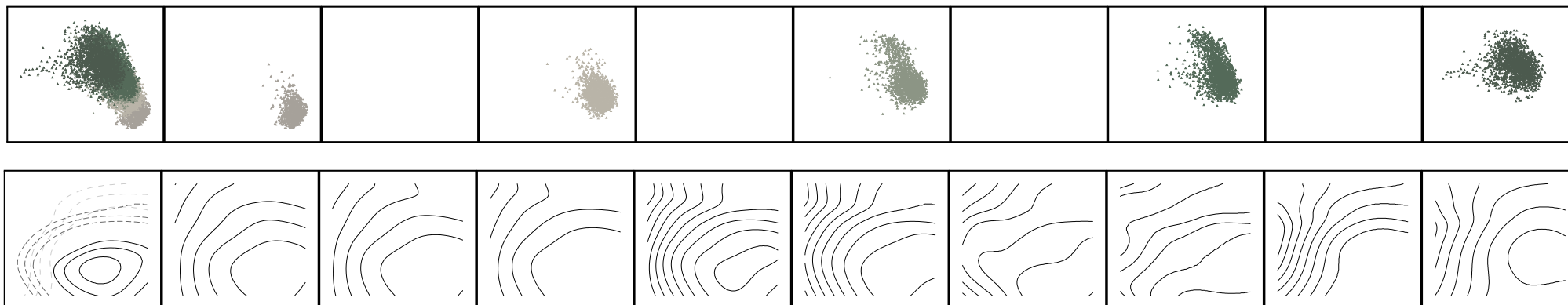
Algorithm 1 JKOnet*: Learning diffusion at lightspeed

- 1: **for** each training step **do**
 - 2: Sample consecutive population snapshots.
 - 3: **Match** particles across snapshots via optimal transport.
 - 4: **Learn**: update the energy by minimizing the MSE between predicted and observed displacements.
 - 5: **end for**
-

results



RNA-seq





separation principles via
optimal transport